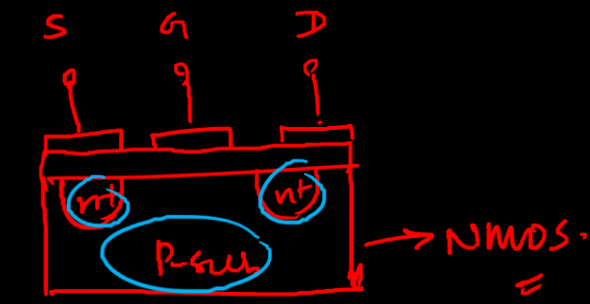
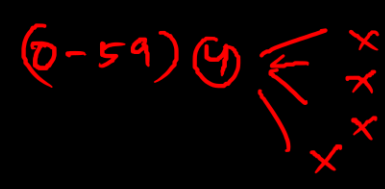
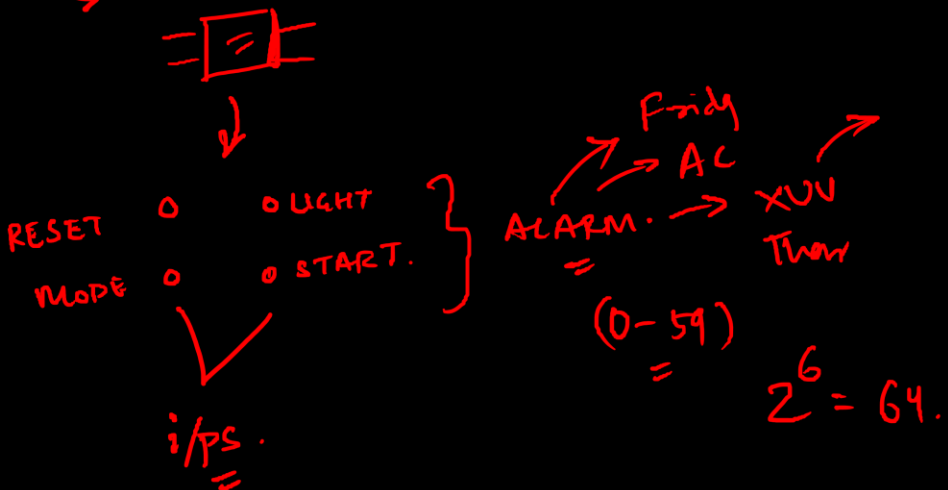
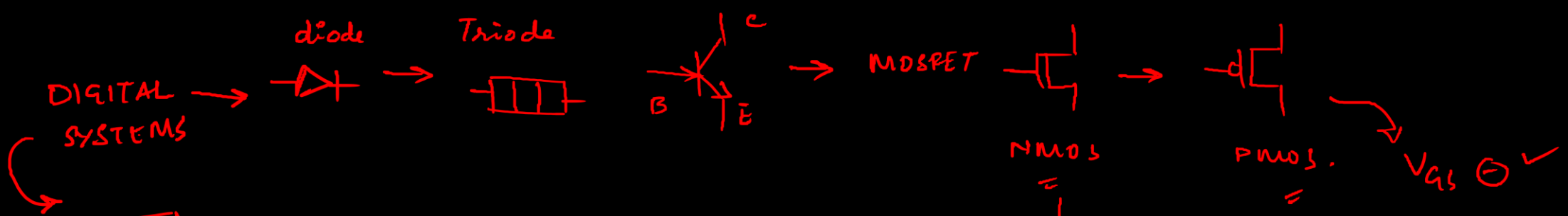


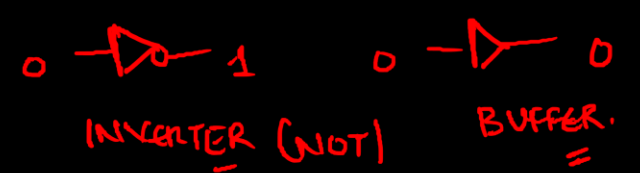
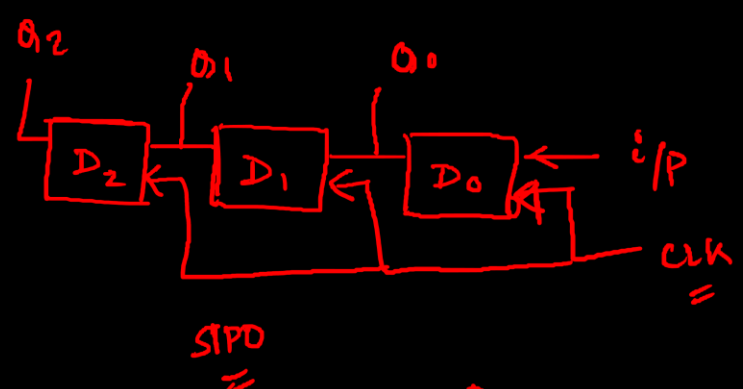
INTERACTION CLASS

17-03-2025



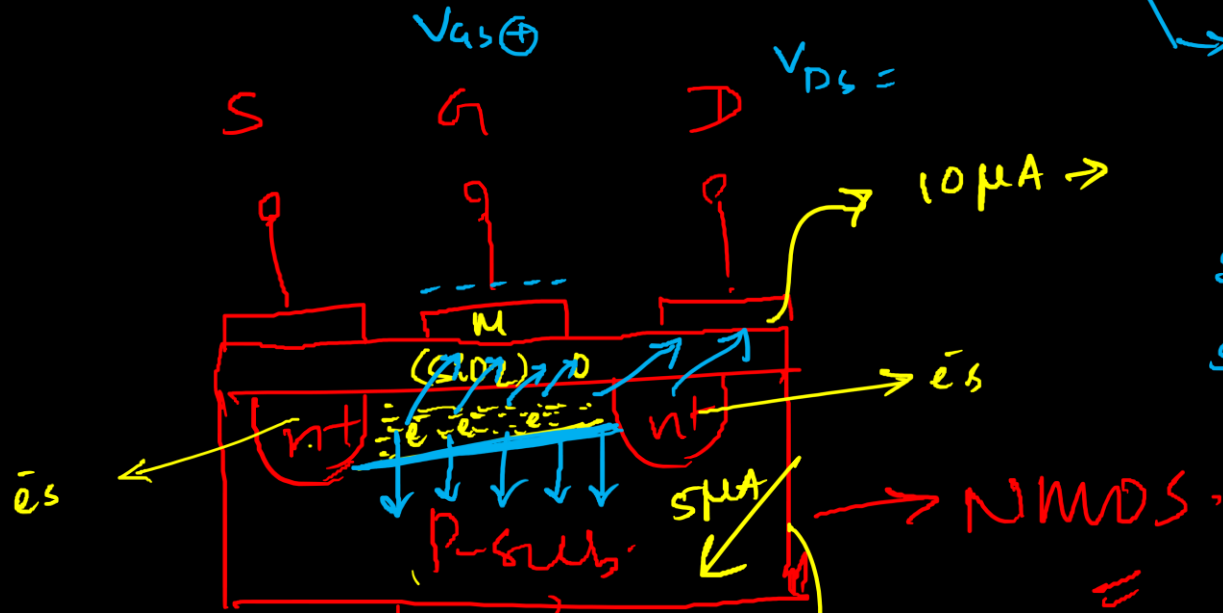
COUNTER → [0 → 7] : 3-bit Counter.

	$\overline{D_2}$	$\overline{D_1}$	$\overline{D_0}$
$2^3 = 8$	0	0	0
0-7	...	...	...
0-(2 ³ -1)	1	1	1



$$V_{GS} = (+ve)$$

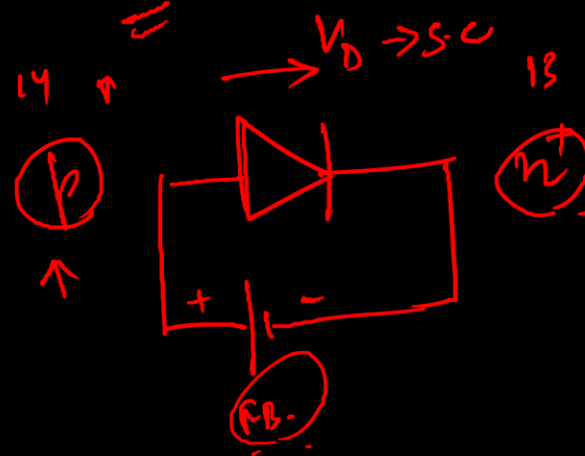
→ ACCUMULATION ; CUT-OFF
 → DEPLETION ; LINEAR
 → INVERSION ; SATURATION.



$S \rightarrow D : \bar{e}_s$
 $S \leftarrow D : \bar{e}_s$



$$1 - 0.7 = 0.3$$



$$\begin{aligned}
 &= 1V \\
 &V_{BE} = 0.7V \\
 &(V_T)_{Si} = 0.7 - 0.6V
 \end{aligned}$$

$$i_c = \frac{dq}{dt}$$

Reverse Saturation Current:

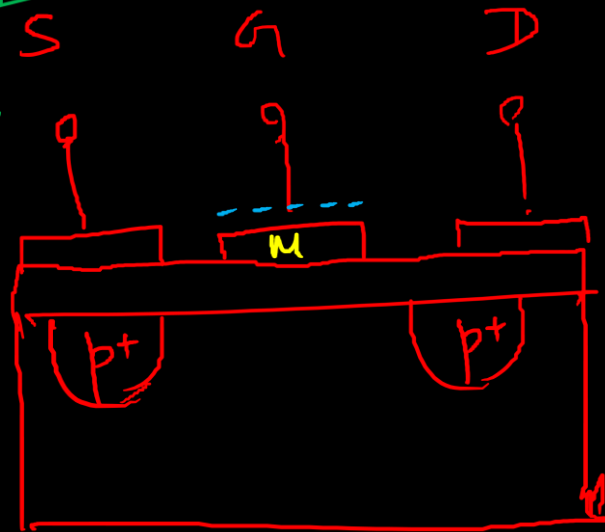
$$I_D = \left(\frac{L}{2}\right) \mu_n C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th}) V_{DS} I_D \sim (V_{GS} - V_{th})$$

$$V_{GS} = (ve)$$

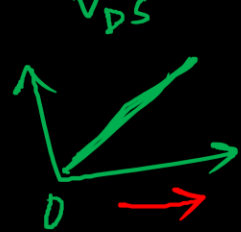
V_{ov} or V_{eff} → ACCUMULATION ; CUT-OFF
 $V_{built-in}$ → DEPLETION ; LINEAR
 → INVERSION ; SATURATION.



$V = IR$
 $I_D \cdot V \propto R_{eff} I$
 $+NM-$
 V_{DS}



PMOS.

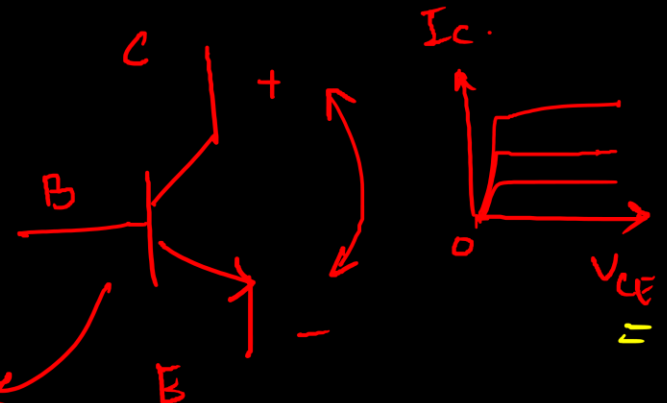


diode → Active → s/c
 $(I_D)_{lin} = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \left\{ (V_{GS} - V_{th}) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right\}$
 $+ve$

CLM.
↑ = W.

$S \rightarrow D : \bar{e}s$
 $S \leftarrow D : \bar{e}$

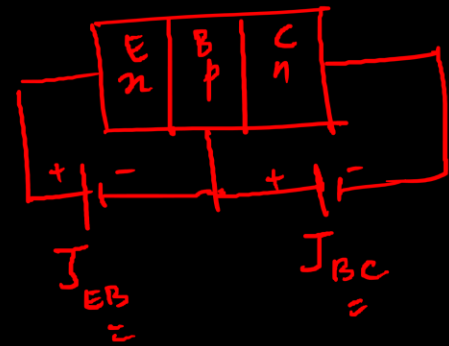
$$V_{CE} = I_C \cdot R_C$$



$I_C = \beta \cdot I_B$
 $\bar{e}/p$

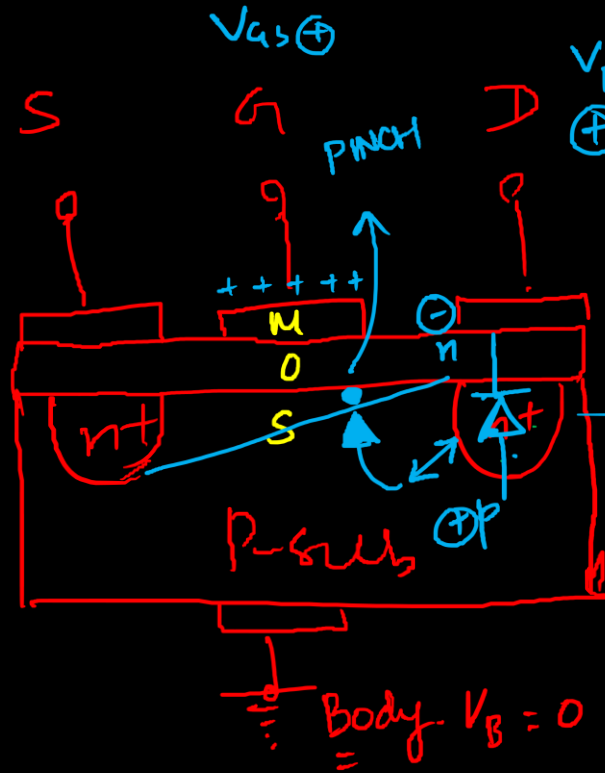
(o/p) current

$$I_E = I_B + I_C$$



$$V_{GS} = (+ve)$$

→ ACCUMULATION ; CUT-OFF
 → DEPLETION ; LINEAR
 → INVERSION ; SATURATION.



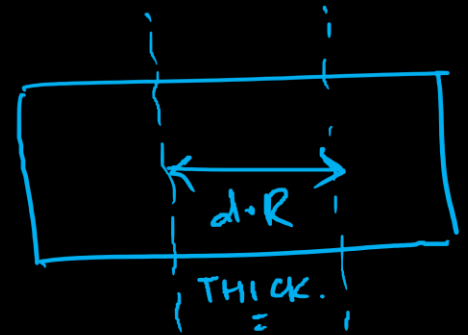
$$V_{DS} = V_D - V_S$$

$$= 8 - 3V$$

$$= 5V$$

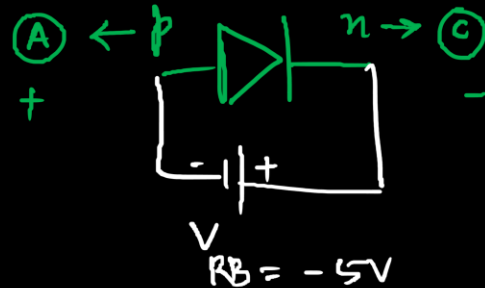
$S \rightarrow D : \bar{e}_s$
 $S \leftarrow D : ?$

→ NMOS.

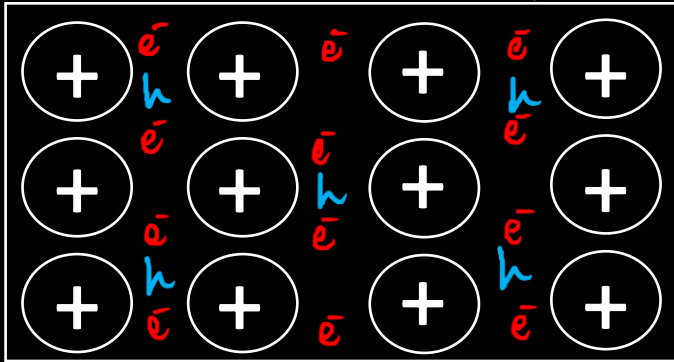


$$I_D = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right] \dots$$

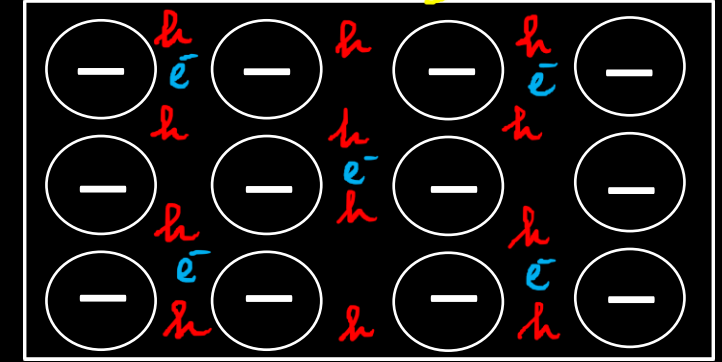
$\lambda = \dots$



Representation of **N**-TYPE s/c



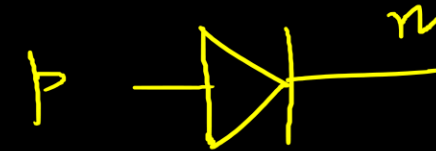
Representation of **P**-TYPE s/c



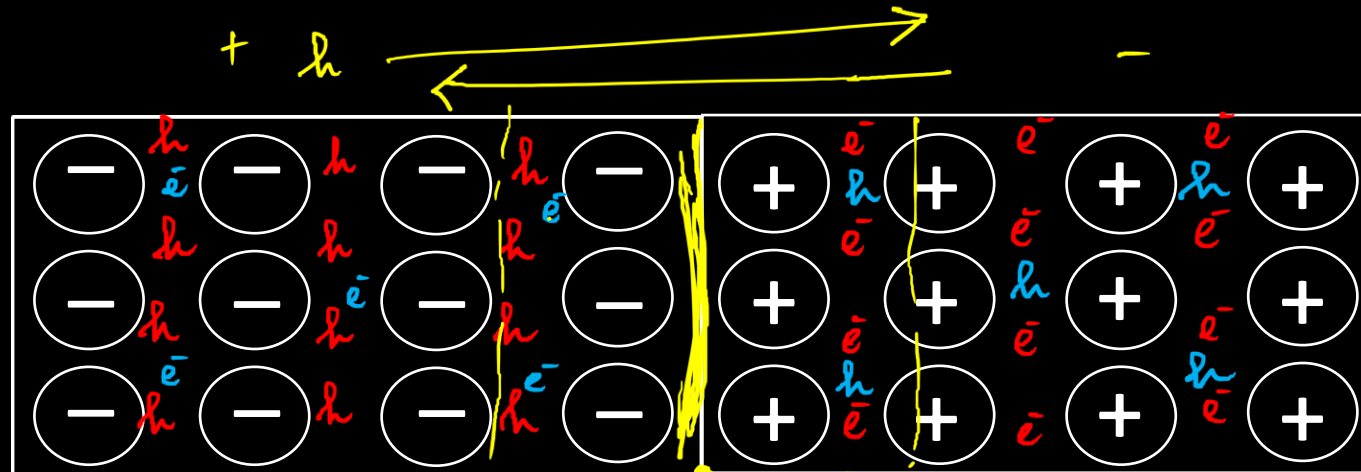
+ = Immobile DONOR ions .

- = Immobile ACCEPTOR ions .

e-s & h = MOBILE .



HOLES MAJ.
e-s minority.



e-s MAJ
holes minority.

P - N Junction